Intake of total, animal and plant protein and subsequent changes in weight or waist circumference in European men and women: the Diogenes project.

BACKGROUND: As protein is considered to increase thermogenesis and satiety more than other macronutrients, it may have beneficial effects on prevention of weight gain and weight maintenance. OBJECTIVE: The objective of this study is to assess the association between the amount and type of dietary protein, and subsequent changes in weight and waist circumference (WC). METHODS: 89,432 men and women from five countries participating in European Prospective Investigation into Cancer and Nutrition (EPIC) were followed for a mean of 6.5 years. Associations between the intake of protein or subgroups of protein (from animal and plant sources) and changes in weight (g per year) or WC (cm per year) were investigated using gender and centre-specific multiple regression analyses. Adjustments were made for other baseline dietary factors, baseline anthropometrics, demographic and lifestyle factors and follow-up time. We used random effect meta-analyses to obtain pooled estimates across centres. RESULTS: Higher intake of total protein, and protein from animal sources was associated with subsequent weight gain for both genders, strongest among women, and the association was mainly attributable to protein from red and processed meat and poultry rather than from fish and dairy sources. There was no overall association between intake of plant protein and subsequent changes in weight. No clear overall associations between intakes of total protein or any of the subgroups and changes in WC were present. The associations showed some heterogeneity between centres, but pooling of estimates was still considered justified. CONCLUSION: A high intake of protein was not found associated with lower weight or waist gain in this observational study. In contrast, protein from food items of animal origin, especially meat and poultry, seemed to be positively associated with long-term weight gain. There were no clear associations for waist changes.